**Background**

Meridian Financial Services Nigeria is a mid-sized financial institution headquartered in Lagos with branch offices in Abuja, Port Harcourt, and Kano. The company employs approximately 5,000 across five departments: Sales, Operations, Risk and Compliance, Human Resources, and Information Technology. Over the past 18 months, the company has experienced a noticeable increase in employee turnover, rising absenteeism, and stagnating performance scores in certain departments. The HR Director has flagged these issues to the Managing Director, and both agree that a data-driven review is overdue.

The objective of this analysis is to examine Meridian's employee data to identify patterns in staff turnover, absenteeism, performance, and compensation across departments and employee demographics. The findings will be used to inform a workforce retention strategy that the HR Director will present to the board.

# HR Workforce Analytics: Understanding increase in employee turnover, rising absenteeism, and stagnating performance scores.

**Tools Used:**

- SQL (Data Cleaning & Analysis)
- Power BI (Dashboard & Visualization)
- MS WORD (Analytical Reports)
- Excel (Initial Data Review and Statistical Test)

This project analyses Meridian's employee workforce data to identify factors influencing employee turnover and retention. The analysis examined employee demographics, salary levels, performance scores, absenteeism patterns, departments, job levels, and employment types.

The findings reveal that employee turnover is concentrated within specific workforce groups. Departments with lower compensation levels, weaker performance indicators, and higher absenteeism recorded increased turnover risk. The analysis also identified high-value employees who demonstrate strong performance and should be prioritized for retention.

Based on these findings, recommendations were developed around compensation review, employee engagement, career development, and targeted retention strategies.

## STATEMENT OF THE PROBLEM

Employee turnover creates significant costs for organizations through recruitment expenses, productivity loss, and knowledge gaps. Meridian requires a data-driven approach to understand why employees leave and identify groups that require retention interventions.

**OBJECTIVES OF THE PROJECT**

The objective of this analysis is to examine Meridian's employee data to identify patterns in staff turnover, absenteeism, performance, and compensation across departments and employee demographics. The findings will be used to inform a workforce retention strategy that the HR Director will present to the board.

The project aims to:

1. Analyse employee turnover patterns.
2. Identify departments with high attrition risk.
3. Evaluate the relationship between salary and retention.
4. Examine the impact of performance and absenteeism on turnover.
5. Identify high-risk and high-value employees.
6. Provide recommendations to improve employee retention.

**RESEARCH QUESTIONS**

1. Which department has the highest employee turnover rate and does it correlate with salary band or length of service?
2. Is there a relationship between absenteeism rate and performance score across departments?
3. Which employee demographics — age group, gender, or employment type — are most associated with voluntary resignation?
4. How does average salary compare across departments and does it align with performance scores?
5. Does education level influence employee exits?
6. Are employees leaving despite receiving competitive salaries?
7. Is salary associated with employee retention?
8. Are high-performing employees being retained?
9. Is poor performance linked to turnover?
10. Does absenteeism predict employee turnover?

**Dataset description**

This dataset contains Meridian HR records of employee information. The dataset includes employee demographics, employee salary and turnover rates. The dataset consists of 5050 records and 17 variables.

**Key Variables**

| Column | Description | Data Type |
|---|---|---|
| Employee_Id | Unique Identify | Categorical |
| Department | Employee Department | Categorical |
| Gender | Employee Gender | Categorical |
| Education_Level | Qualification | Categorical |
| Employment_Type | Mode of Employment | Categorical |
| Job_Level | Employee Ranks | Categorical |
| Monthy_Salary_Ngn | Employee Salary | Numerical |
| Performance_Score | Employee Performance Rating | Numerical |
| Absenteeism_Days | Days Absent | Numerical |
| Turnover_Status | Active/Exited Status | Categorical |

**DATA CLEANING PROCESS**

| **Data quality issues** | **Data cleaning steps** |
|---|---|
| Duplicated records | Removed all duplicates, |
| Column Names are in small letters | Capitalise each word |
| The gender column has missing values | Populate it with gender of customers that has the same first name |
| Inconsistencies in the gender column | Standardized to Sentences Case |
| Department column has missing values | Filled with unknown. |
| Employment type has missing values | Filled with unknown |
| Inconsistencies in Employment Type | Standardized to Sentences Case |
| Age group has missing value | Populated with the mode (most occurred age group 31-40) because it is a categorical data. |
| Missing Values in Job, Education Level | Filled with unknown |
| Exit Date has missing values | Filled it with unknown. |
| Missing values in monthly salary column | Filled with average monthly salary |
| Missing values in performance column | Filled with average performance score (3.42) |
| Missing values in absenteeism days | Filled with average absenteeism days(10days) |

**Note:** Performance scores ranged from -0.50 to 5.00, with an average score of 3.43. The slight negative minimum was retained because the dataset appears to use a continuous performance index rather than a strict 1–5 rating scale. The average performance score indicates that employees generally performed above the midpoint of the scale.

**Created New Variables:** Salary Band and Risk Category

## METHODOLOGY

The tools use for this project are:

**SQL** was used for data cleaning, transformation, querying and descriptive HR analysis.

**Excel** was used for statistical testing, including correlation, ANOVA and chi-square analysis.

**Power BI** was used for interactive visualization and dashboard development.

**Microsoft Word** was used to document the methodology, findings, conclusions and recommendations.

## STATISTICAL ANALYSIS

### Overview

A statistical analysis was conducted on the Meridian employee dataset containing 5,000 employee records. The purpose of the analysis was to identify statistically significant relationships and differences among key human resource variables, particularly employee turnover, salary, performance, absenteeism, and department. The analysis combined descriptive and inferential statistical techniques. Descriptive statistics were used to summarize the characteristics of the dataset, while inferential statistical tests were applied to determine whether observed differences and relationships were statistically significant.

The statistical analyses were conducted using a significance level (α) of 0.05. Therefore, results with a p-value below 0.05 were considered statistically significant, while results with a p-value greater than 0.05 were considered not statistically significant.

The major statistical techniques used were:

- t-tests to compare salary, performance, and absenteeism between employees who
- remained with the organization and employees who left.
- One-way ANOVA to examine differences in salary, absenteeism and performance across departments.
- Chi-square tests of independence to examine the association between categorical variables such as department and turnover status.
- Pearson correlation analysis to examine relationships among salary, performance, and absenteeism.

**DESCRIPTIVE ANALYSIS**

**Workforce composition:** There are 5000 employees in meridian hr dataset, spread across 5 departments: sales, operations, human resources, risk and compliance and information technology. Operations have the largest rate of employees (23.98%) Followed closely by Sales (23.30%), Risk and compliance recorded 19.66%, Information Technology recorded 14.44% and Human Resources recorded 14.42%. The company recruit workers with different qualifications: BSc, MSc, HND OND PhD; BSc. Is the qualification with the highest employees with 38.76% of the workforce and PhD is the lowest.

**Turnover:** 1790 employees out of the 5000 employed left. The department affected most by turnover is sales department with almost half of its entire workforce either resign, retire or got terminated. PhD is the Education level with highest turnover and contract employees has the highest turnover rate.

**Salary:** Monthly salary ranges from 85,000 – 2,200,000. The expected monthly pay (average salary) of each employee is approximately 357000 naira. Sales department is the lowest paid departments. Sales Department average take home is way below company overall average salary. Junior and mid-levels staff are low salary earners. The employees are group into different salary band and low salary band is the highest salary (between 100000 and 257000) band in the workforce. This means majority of the workforce receive low salaries. Generally, salary does not have a significant relationship with turnover.

**Performance:** the average overall performance is 3.43. Sales department has the highest number of employees with performance scores below 3.4. junior level employees are the employee level with the lowest performance. Performance has a significant relationship with turnover.

**Absenteeism:** absenteeism is spread across department, job and education level with operations recording the highest absenteeism days. Absenteeism does not significantly predict turnover.

**STATISTICAL ANALYSIS**

**Salary vs Turnover**

Does mean salary differ between employees who remained and those who left?

**$H_0$:** There is no significant difference in mean salary between employees who remained and employees who left.

**$H_1$:** There is a significant difference in mean salary between the two groups.

| | *Active Salary* | *Turnover Salary* |
|---|---|---|
| Mean | 359145.7306 | 353167.8278 |
| Variance | 78685837224 | 71850676847 |
| Observations | 3210 | 1790 |
| Hypothesized Mean Difference | 0 | |
| df | 3842 | |
| t Stat | 0.743456051 | |
| P(T<=t) one-tail | 0.22862554 | |
| t Critical one-tail | 1.645250331 | |
| P(T<=t) two-tail | 0.45725108 | |
| t Critical two-tail | 1.960581633 | |

t-test was conducted to examine whether employee salary differed across turnover-status groups. The analysis was done by grouping salary into active salary and turnover salary. The analysis produced a p-value of 0.45725, which is above the 0.05 significance level. Therefore, the null hypothesis was not rejected (fail to reject), indicating that There is no significant difference in mean salary between employees who remained and employees who left.

**Conclusion:** There is no statistically significant relationship between salary and turnover

**Performance and turnover**

Does mean performance differ between employees who remained and those who left?

**$H_0$:** There is no significant difference in mean performance between employees who remained and employees who left.

**$H_1$:** There is a significant difference in mean performance between the two groups.

| | *active - performance* | *turnover-performance* |
|---|---|---|
| Mean | 3.47123676 | 3.364553073 |
| Variance | 0.766252225 | 0.79825008 |
| Observations | 3210 | 1790 |
| Hypothesized Mean Difference | 0 | |
| df | 3636 | |
| t Stat | 4.07719323 | |
| P(T<=t) one-tail | 2.32791E-05 | |
| t Critical one-tail | 1.645272813 | |
| P(T<=t) two-tail | 4.65582E-05 | |
| t Critical two-tail | 1.960616638 | |

T-test was conducted to examine whether employee performance differed across turnover-status groups. The analysis produced a p-value of 0.0000466, which is below the 0.05 significance level. Therefore, the null hypothesis was rejected, indicating that mean performance scores differ significantly across turnover-status groups.

**Conclusion:** There is statistically significant relationship between performance and turnover

**Absenteeism and turnover**

Does mean absenteeism differ between employees who remained and those who left?

**$H_o$:** There is no significant difference in mean absenteeism between employees who remained and employees who left.

**$H_1$:** There is a significant difference in mean absenteeism between the two groups.

| | *active-turnover* | *turnover-absenteeism* |
|---|---|---|
| Mean | 9.970093458 | 10.41284916 |
| Variance | 71.37897444 | 83.81213218 |
| Observations | 3210 | 1790 |
| Hypothesized Mean Difference | 0 | |
| df | 3457 | |
| t Stat | -1.684823688 | |
| P(T<=t) one-tail | 0.046056428 | |
| t Critical one-tail | 1.645294524 | |
| P(T<=t) two-tail | 0.092112855 | |
| t Critical two-tail | 1.960650443 | |

t-test was conducted to examine whether absenteeism differed across turnover-status groups. The analysis was done by grouping absenteeism into active absenteeism and turnover absenteeism. The analysis produced a p-value of 0.0921, which is above the 0.05 significance level. Therefore, the null hypothesis was not rejected (fail to reject), indicating that There is no significant difference in mean absenteeism between employees who remained and employees who left.

**Conclusion:** There is no statistically significant relationship between absenteeism and turnover

**ANOVA**

| Test | f-statistic | p-value | decision |
|---|---|---|---|
| Salary by department | 37.58573 | 5.3E-31 | significant |
| Performance by department | 88.40088 | 1.38E-72 | significant |
| Absenteeism by department | 0.412341 | 0.799883 | Not significant |

**Salary and department**

Does mean salary differ across departments?

**$H_o$:** Mean salary is equal across departments.

**$H_1$:** At least one department has a different mean salary.

A one-way ANOVA was conducted to examine whether employee mean salary differed across department. The analysis produced a p-value of 5.3E-31, which is below the 0.05 significance level. Therefore, the null hypothesis was rejected, indicating that mean salary differs significantly across departments.

**Performance and department**

Does mean performance differ across departments?

**$H_o$:** Mean performance is equal across departments.

**$H_1$:** At least one department has a different mean performance.

ANOVA was conducted to examine whether employee mean performance differed across department. The analysis produced a p-value of 1.38E-72, which is below the 0.05 significance level. Therefore, the null hypothesis was rejected, indicating that mean performance differs significantly across departments.

**Absenteeism and department**

Does mean absenteeism differ across departments?

**$H_o$:** There is no statistically significant difference in mean absenteeism across departments.

**$H_1$:** There is a statistically significant difference in mean absenteeism across departments.

ANOVA was conducted to examine whether employee mean absenteeism differed across department. The analysis produced a p-value of 0.799883, which is above the 0.05 significance level. Therefore, the null hypothesis was not rejected (fail to reject), indicating that there is no significant difference in mean absenteeism across departments.

**CHI-SQUARE ANALYSIS**

**Department and turnover**

Is department significantly associated with turnover?

**$H_o$:** Department and turnover status are independent.

**$H_1$:** Department and turnover status are associated.

| Row Labels | Active | Left | total | expected active | expected left | p-value |
|---|---|---|---|---|---|---|
| Human Resources | 486 | 235 | 721 | 462.882 | 258.118 | 9.0175E-27 |
| Information Technology | 474 | 248 | 722 | 463.524 | 258.476 | |
| Operations | 791 | 408 | 1199 | 769.758 | 429.242 | |
| Risk and Compliance | 722 | 261 | 983 | 631.086 | 351.914 | |
| Sales | 594 | 571 | 1165 | 747.93 | 417.07 | |
| Unknown | 143 | 67 | 210 | 134.82 | 75.18 | |
| **Grand Total** | **3210** | **1790** | **5000** | 3210 | 1790 | |

The Chi-square test indicates a statistically significant association between department and turnover status ($p = 9.0175 \times 10^{-27}$). Therefore, the null association

was rejected, turnover status is not independent of department, suggesting that employee turnover patterns differ significantly across departments.

Statistical analysis indicates that department membership is significantly associated with turnover, with Sales showing higher employee exit patterns compared with other departments.

**CORRELATION ANALYSIS**

| Variables | Pearson r | p-value | Interpretation |
|---|---|---|---|
| Salary & Performance | 0.24309 | 3.62E-68 | Significant |
| Salary & Absenteeism | 0.01246 | 0.3783 | Not significant |
| Performance & Absenteeism | 0.01177 | 0.4054 | Not significant |

**Salary and Performance**

Is salary significantly correlated with performance

**$H_0$:** There is no significant relationship between employee salary and performance score.

**$H_1$:** There is a significant relationship between employee salary and performance score.

The correlation analysis shows that the p-value is 3.62E-68 which is below the 0.05 significant level. This is an extremely small p-value. Therefore, null hypothesis was rejected. There is statistically significant evidence of a relationship between salary and performance. However, correlation coefficient (r) = 0.24309 suggest a very weak positive relationship

**Salary and absenteeism**

Is salary significantly correlated with absenteeism?

**$H_0$:** There is no significant relationship between employee salary and absenteeism**.**

**$H_1$:** There is a significant relationship between employee salary and absenteeism**.**

The correlation analysis shows that the p-value is 0.3783, which is above the 0.05 significant level. Therefore, null hypothesis was not rejected (fail to reject). There is no significant evidence of a relationship between salary and absenteeism.

So, there is not enough statistical evidence of a linear relationship between salary and absenteeism.

**Performance and Absenteeism**

Is performance significantly correlated with absenteeism?

**$H_0$:** There is no significant relationship between employee performance and absenteeism.

**$H_1$:** There is a significant relationship between employee performance and absenteeism.

The correlation analysis shows that the p-value is 0.4054, which is above the 0.05 significant level. Therefore, null hypothesis was not rejected (fail to reject). There is no significant evidence of a relationship between performance and absenteeism.

There is not enough statistical evidence of a linear relationship between performance and absenteeism.

Correlation analysis showed a statistically significant relationship between salary and performance ($p < 0.001$). However, salary was not significantly related to absenteeism ($p = 0.378$), and performance was not significantly related to absenteeism ($p = 0.405$). This suggests that, within the dataset, salary and performance are statistically associated, while absenteeism does not show a statistically significant linear association with either salary or performance.

The correlation analysis indicates that salary has a statistically significant but weak positive relationship with employee performance ($r = 0.243$, $p < 0.001$). However, salary has virtually no relationship with absenteeism ($r = 0.012$, $p = 0.378$), while performance also has virtually no relationship with absenteeism ($r = 0.012$, $p = 0.405$). Therefore, higher salaries are associated with slightly higher performance, but neither salary nor performance appears to be a meaningful linear predictor of absenteeism in the dataset.

**POWER BI DASHBOARD ANALYSIS**

The Power BI dashboard was developed to provide an interactive visualization of the Meridian HR dataset. The dashboard allows users to examine workforce composition, turnover, salary, performance and absenteeism across different employee characteristics. Slicers were added to allow users to filter the analysis by department, gender, education level, employment type, job level and salary band.

The Power BI contains pages of dashboards highlighting visually the trends and patterns found in the meridian hr dataset.

The pages are:

- ✓ Workforce overview analysis
- ✓ Turnover analysis
- ✓ Salary analysis
- ✓ Performance and absenteeism analysis
- ✓ Employee risk profile

**WORKFORCE ANALYSIS**

Figure 1: WORKFORCE DASHBOARD

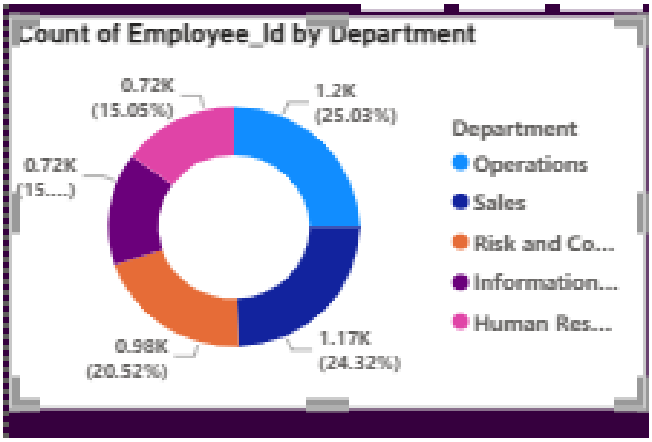


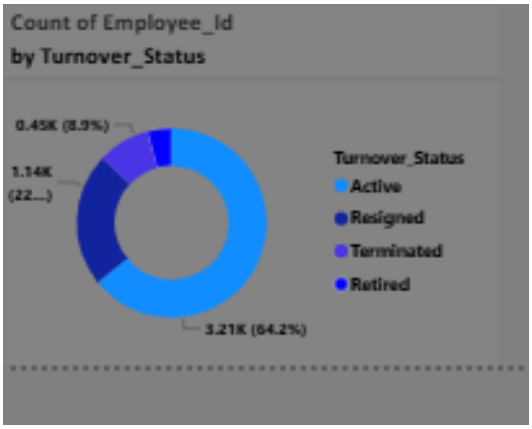

Figure 1 illustrate the executive dashboard; it provides an overall view of the Meridian workforce. It summarizes the total employee population according to employee demography. The meridian workforce comprises of 5,000 employees, spread across 5 department: sales, human resources, operations, information technology and risk and compliance. Operation has the largest workforce 25.03% and human resources is the department with the lowest workforce.

**TURNOVER ANALYSIS**

Figure 2




Figure 3




The turnover dashboard shows differences in employee retention across departments and employee characteristics. The department comparison highlights departments with relatively higher and lower turnover rates. Additional breakdowns by education level, employment type, job level and salary band allow the analysis to identify employee groups experiencing higher turnover.

As shown in figure 2, The department with the highest turnover rate 49.01% is sales, while risk and compliance recorded the lowest turnover rate of 26.55%. This suggests that employee retention is not uniform across the organization and that retention interventions may need to be targeted toward specific departments.

Figure 3 shows the rate of turnover status of the workforce. 64.2% of the workforce is active, 22.74% resigned, 8.9% got terminated and 4.16% retired. 22.74% resigned, this suggests that there are some underlying issues making employee to resign voluntarily.

**SALARY ANALYSIS**

Figure 4




Figure 5




Figure 6




Figure 4 shows that risk and compliance recorded the highest Average salary while sales recorded the lowest. In figure 5, The employees were group into salary bands, chart illustrates that low salary band is the highest employees in the dataset and poor salary band has the highest turnover.

However, in figure 6, when salary was compared with turnover, the comparison indicates that salary alone does not necessarily correspond to lower turnover. For example, Information Technology has a relatively high average salary but records a relatively high turnover rate, while human resources that has an average salary that is 33,000 lower than information technology recorded a lower turnover rate. This suggests that factors other than compensation may contribute to employee turnover, such as workload, career progression, management, job satisfaction or working conditions. However, these potential explanations cannot be confirmed from the available dataset and would require additional variables or qualitative research.

**PERFORMANCE AND ABSENTEEISM**

Figure 7

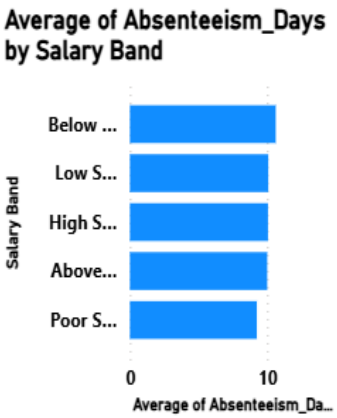



Figure 8

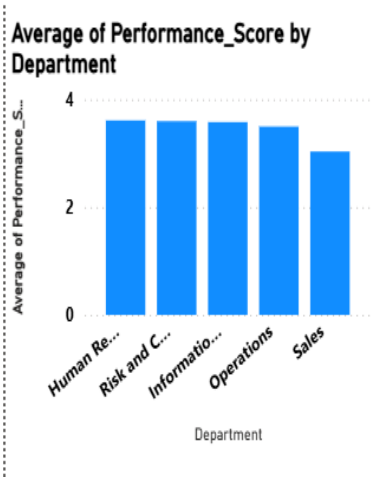
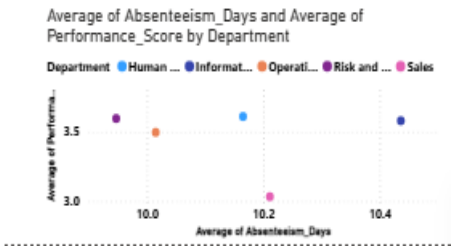



Figure 9




The performance and absenteeism dashboard compares employee performance and absenteeism across departments, salary bands and job levels. The analysis helps identify departments with relatively high, low and average performance and examines whether performance patterns vary according to compensation and employee status. The visualization helps identify employee groups with relatively high absenteeism and allows absenteeism patterns to be compared with performance and turnover.

As shown in figure 7, below average salary earners recorded average absenteeism days of 10.63, low salary recorded 10.08, high salary recorded 10.06 while poor salary recorded the least average absenteeism days of 9.25. this suggests that there is no significant relationship between absenteeism and salary.

Figure 8 shows the relationship between performance and department. Human resources is the department with highest average performance while sales is the lowest. This means performance is not the same across department and intervention strategies should be more targeted toward some department than others.

Figure 8 shows the relationship between performance and absenteeism by department. Risk and compliance recorded average performance of 3.60 and average absenteeism of 9.95 days while Human resources recorded 3.61 average performance and 10.17 average absenteeism days. The scatter plots indicate no linear relationship between performance and absenteeism.

The descriptive analysis suggests a relationship between performance and salary, and no relationship between absenteeism, salary and performance. This relationship was subsequently examined statistically using correlation analysis in Excel to determine whether the observed relationship was statistically significant.

**EMPLOYEE RISK PROFILE**

Figure 10

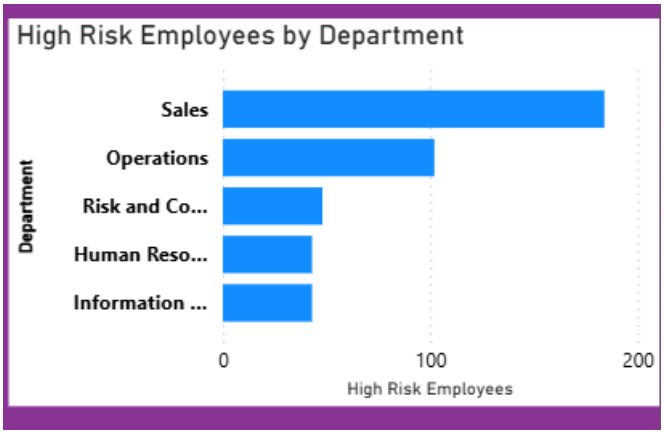



Figure 10 illustrate the risk profile of the workforce. Sales department has the largest high risk employees while human resources and information technology recorded the smallest.

**KEY DASHBOARD INSIGHTS**

**Turnover concentration:** some departments experience higher turnover rates than others.

**Compensation:** compensation differ substantially across departments and job levels.

**Salary and retention:** Higher salary does not necessarily correspond to lower turnover.

**Performance:** Performance varies across departments and employee groups.
**Absenteeism:** Absenteeism patterns differ across employee groups.

**Employee characteristics:** Turnover varies across selected demographic and employment categories.

**KEY ANALYSIS FINDINGS**

**Finding 1: Workforce Overview**

The organization consists of employees across multiple departments, employment type and job levels. The company employed with different categories of qualifications spread across different age bracket. Analysis of workforce distribution shows that Operations has the largest employee population, Full-time is the largest employment type, BSc. Makes the largest qualifications and Mid-level is the largest job-level in the workforce, making it important when evaluating workforce trends.

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**Finding 2: Turnover Analysis**

Dashboard illustrate sales as the department with the highest turnover rate of 49.01%, while risk and compliance recorded the lowest turnover rate of 26.55%.

The Chi-square test indicates a statistically significant association between department and turnover status ($p = 9.0175 \times 10^{-27}$).

Statistical analysis indicates that department membership is significantly associated with turnover, with Sales showing higher employee exit patterns compared with other departments.

Turnover analysis revealed that employee exits are not evenly distributed across departments and all other variables in this organization and that retention strategies may need to be targeted rather than applied uniformly across the organization. Sales recorded the highest turnover, suggesting potential challenges related to compensation, workload, or employee satisfaction.

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**Finding 3: Salary Analysis**

Risk and compliance recorded the highest Average salary while sales recorded the lowest.

A one-way ANOVA was conducted to examine whether employee mean salary differed across department. The analysis produced a p-value of 5.3E-31, which is below the 0.05 significance level, indicating that there is a statistically significant difference in mean salary across departments.

Also, t-test was conducted to examine whether employee salary differed across turnover-status groups. The analysis was done by grouping salary into active salary

and turnover salary. The analysis produced a p-value of 0.45725, which is above the 0.05 significance level, indicating that There is no statistically significant relationship between salary and turnover

Salary analysis showed differences in compensation across departments and other variables. Although higher salaries generally support retention, turnover patterns indicate that compensation alone does not fully explain employee exits.

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**Finding 4: Performance Analysis**

Human resources is the department with highest average performance while sales is the lowest. This means performance is not the same across department and intervention strategies should be more targeted toward some department than others.

ANOVA was conducted to examine whether employee mean performance differed across department. The analysis produced a p-value of 1.38E-72, which is below the 0.05 significance level. Therefore, the null hypothesis was rejected, indicating that mean performance differs significantly across departments.

Also, T-test was conducted to examine whether employee performance differed across turnover-status groups. The analysis produced a p-value of 0.0000466, which is below the 0.05 significance level, indicating that there is statistically significant relationship between performance and turnover

And The correlation analysis shows that the p-value is 3.62E-68 which is below the 0.05 significant level. This is an extremely small p-value. Therefore, null hypothesis was rejected. There is statistically significant evidence of a relationship between salary and performance. However, correlation coefficient (r) = 0.24309 suggest a very weak positive relationship

Performance analysis showed variation across departments. Some departments maintained strong performance levels while experiencing lower turnover. Also, there is evidence of relationship between performance and salary, suggesting that employee engagement, compensation review and working conditions may influence productivity.

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**Finding 5: Absenteeism Analysis**

The dashboard shows below average salary earners recorded average absenteeism days of 10.63, low salary recorded 10.08, high salary recorded 10.06 while poor salary recorded the least average absenteeism days of 9.25.

t-test was conducted to examine whether absenteeism differed across turnover-status groups. The analysis was done by grouping absenteeism into active absenteeism and turnover absenteeism. The analysis produced a p-value of 0.0921, which is above the 0.05 significance level, indicating that There is no statistically significant relationship between absenteeism and turnover

ANOVA was conducted to examine whether employee mean absenteeism differed across department. The analysis produced a p-value of 0.799883, which is above the 0.05 significance level., indicating that mean absenteeism does not differs significantly across departments.

The correlation analysis shows that the p-value is 0.3783, which is above the 0.05 significant level. Therefore, there is no significant evidence of a relationship between salary and absenteeism.

Also, correlation analysis shows that the p-value is 0.4054, which is above the 0.05 significant level. Therefore, there is no significant evidence of a relationship between performance and absenteeism.

Absenteeism is spread across all department and other variable in the dataset. Compensation, performance, turnover does not influence absenteeism. Departments with higher absenteeism levels require further investigation because frequent absence may indicate low engagement, dissatisfaction, workload challenges, or workplace issues.

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**Finding 6: Employee Risk Segmentation**

Risk scoring identified employees who combine multiple turnover indicators, including low salary, low performance, and high absenteeism. These employees represent priority groups for HR intervention. Sales department recorded the highest proportion of high-risk employees, driven by lower salary levels, higher absenteeism, and weaker performance indicators. This suggests the department requires targeted retention interventions."

| Department | Salary | Performance | Absenteeism | Turnover Risk |
|---|---|---|---|---|
| Sales | Low | Low | High | High |
| Operations | Below Average | Good | Low | Medium |
| IT | High | Good | High | Medium |
| HR | Above Average | Highest | Low | Low |
| Risk & Compliance | Highest | High | Lowest | Low |

In conclusion, the analysis revealed that turnover is concentrated in specific employee groups and departments. Sales experienced the highest workforce risk due to lower compensation, weaker performance indicators, and higher absenteeism. Risk & Compliance demonstrated stronger retention patterns due to competitive salaries and better employee outcomes. Salary appears to influence retention, but factors such as attendance, performance, and employment type also contribute to employee turnover.

**RECOMMENDATIONS**

Improve Retention

- Review salary structure.
- Investigate workload challenges.
- Create career growth opportunities.
- Improve employee engagement.

Improve performance

- Recognition programs.
- Promotion opportunities.
- Competitive rewards.

Reduce Absenteeism

- Identify causes of frequent absence.
- Improve workplace conditions.
- Introduce employee wellbeing initiatives.
- Introduce health insurance for the workforce.

Review Contract Employment

- Provide clearer contract renewal policies.
- Improve job security.
- Create career progression pathways.

The analysis of Meridian's workforce data identified employee turnover patterns linked to compensation, department characteristics, performance, and absenteeism. Sales emerged as the highest-risk department due to lower salary levels, weaker performance indicators, and increased absenteeism. Risk & Compliance demonstrated stronger retention outcomes, supported by competitive compensation and employee performance.

To improve retention, management should prioritize salary reviews, employee development programs, attendance improvement initiatives, and targeted engagement strategies for high-risk groups.

**LIMITATIONS**

- The dataset contains 5,000 records.
- The analysis is based on available variables.
- Correlation does not establish causation.
- Turnover may be influenced by factors not included in the dataset.
- Statistical significance does not necessarily imply practical/business significance.
- Some variables may require additional organizational context.

## CONCLUSION

The Meridian HR Analytics project demonstrates how data can be used to understand workforce behaviour and improve decision-making. Employee turnover is influenced by multiple factors including compensation, performance, absenteeism, and employment conditions. By identifying high-risk groups and understanding departmental challenges, HR can implement targeted strategies to reduce turnover, reduce absenteeism and maintain high organizational performance.